

CAPE ROSS, Antarctica

WMO: 896660

Lat: 76.715S

Lon: 162.970E

Elev: 151

StdP: 99.52

Time Zone: 12.00 (NZL)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-34.6	-33.3	N/A	N/A	N/A	N/A	N/A	N/A	23.5	-17.0	20.8	-19.9	5.7	170	1.294

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.2	2.1	N/A	1.0	N/A	0.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	17.2	14.6	DB	-37.2	4.4	2.1	1.6	-38.7	5.6	-39.9	6.5	-41.0	7.4	-42.5	8.6	
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-15.4	-2.5	-7.8	-14.9	-19.5	-20.4	-21.4	-24.5	-22.6	-21.8	-16.7	-9.0	-3.3
	DBStd	8.71	2.07	3.02	3.93	4.79	5.66	5.32	5.58	5.74	5.43	3.72	3.88	2.28
	HDD10.0	9272	388	499	771	886	943	942	1069	1010	954	829	570	411
	HDD18.3	12313	646	733	1029	1136	1202	1192	1327	1268	1204	1087	820	669
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.3	4.6	6.8	7.1	6.6	6.8	7.4	6.4	7.3	7.3	6.2	5.1	4.4
Precipitation	PrecAvg	212	23	24	17	13	16	19	13	19	17	17	17	17
	PrecMax	469	49	60	69	35	48	62	37	73	58	61	52	62
	PrecMin	47	2	1	1	5	1	2	0	0	0	0	1	2
	PrecStd	108	16	17	15	7	13	14	9	18	13	15	11	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	-0.3	-5.6	-5.6	-6.9	-7.9	-8.5	-8.2	-8.5	-4.7	-0.4	3.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	2.2	-1.8	-7.1	-9.2	-9.2	-10.0	-11.3	-10.8	-10.6	-8.6	-1.9	2.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	1.2	-2.8	-8.3	-10.9	-10.8	-11.9	-13.9	-12.9	-12.3	-10.1	-2.8	1.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	0.4	-3.9	-9.7	-12.9	-12.5	-13.9	-16.2	-14.5	-14.2	-11.4	-3.9	0.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	3.2	3.3	3.5	4.5	4.8	5.1	5.5	6.1	5.7	4.8	4.3	3.6
		MCDBR	4.1	4.1	4.6	5.9	5.8	6.6	7.3	7.2	7.1	6.0	5.2	4.8
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.236	0.224	0.179	0.111	N/A	N/A	N/A	0.109	0.167	0.197	0.218	0.234
	taud		2.305	2.320	2.229	1.906	N/A	N/A	N/A	1.888	2.172	2.278	2.295	2.302
	Ebn at Noon		975	906	791	237	0	0	0	216	793	942	996	999
	Edh at Noon		95	74	48	7	0	0	0	6	48	76	95	101
All-Sky Solar Radiation	RadAvg		8.38	4.97	1.84	0.19	0.00	0.00	0.00	0.04	1.04	3.89	7.71	9.65
	RadStd		0.20	0.16	0.08	0.01	0.00	0.00	0.00	0.00	0.05	0.12	0.20	0.25

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (4 neighbors)	+0.80	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page